Question no 1:

Step 1: Symbolize the propositions

1. **"Alice is a manager"**
   Let M represent "Alice is a manager."
2. **"Bob is a supervisor"**
   Let S represent "Bob is a supervisor."

Step 2: Translate the premises into logical statements

1. **Premise 1: "If Alice is a manager, then Bob is not a supervisor."**
   This is a conditional statement:
   $M \rightarrow \neg S$
   (If Alice is a manager, then Bob is not a supervisor.)
2. **Premise 2: "If Bob is a supervisor, then Alice is not a manager."**
   Another conditional statement:
   $S \rightarrow \neg M$
   (If Bob is a supervisor, then Alice is not a manager.)

Step 3: Translate the conclusion

The conclusion is:
"**Alice is not a manager, and Bob is not a supervisor.**"
This is a conjunction: $\neg M \land \neg S$

Step 4: Write the argument formally

The argument can now be written as:

1. $M \rightarrow \neg S$ (Premise 1)
2. $S \rightarrow \neg M$ (Premise 2)
3. $\neg M \land \neg S$ (Conclusion)

We need to check whether the conclusion follows logically from the premises.

Step 5: Construct a truth table

We will list all possible truth values for M (Alice being a manager) and S (Bob being a supervisor) and evaluate the premises and the conclusion.

M	S	$\neg S$	$M\rightarrow\neg S$	$S\rightarrow\neg M$	$\neg M$	$\neg S$	$\neg M\wedge\neg S$
T	T	F	F	F	F	F	F
T	F	T	T	T	F	T	F
F	T	F	T	T	T	F	F
F	F	T	T	T	T	T	T

Step 6: Evaluate the truth table

- **Row 3**: In this case, $M \rightarrow \neg S$ and $S \rightarrow \neg M$ are both true. The conclusion $\neg M \wedge \neg S$ is also true in this row. So, the argument holds true in this case.
- **Other rows**: In the other rows, at least one of the premises is false. For example:
  - **Row 1**: Both premises are false, so we don't need to check the conclusion.
  - **Row 2**: The first premise is true, but the second one is false.
  - **Row 4**: The second premise is true, but the first one is false.

Step 7: Conclusion

The argument is **valid** because in the case where both premises are true (row 3), the conclusion is also true. Therefore, the argument is logically valid based on the truth table.

Question no 2:

To prove $A\backslash B = A \cap B^c$

1. Show $A\backslash B \subseteq A \cap B^c$:

If $x \in A\backslash B$, then $x \in A$ and $x \notin B$

So, $x \in B^c$

Thus, $x \in A \cap B^c$

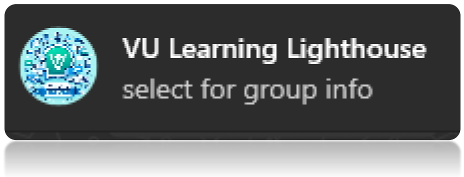
2. Show $A \cap B^c \subseteq A\backslash B$:

If $x \in A \cap B^c$, then $x \in A$ and $x \in B^c$

So, $x \notin B$

Thus, $x \in A\backslash B$

Conclusion: $A\backslash B= A \cap B^c$



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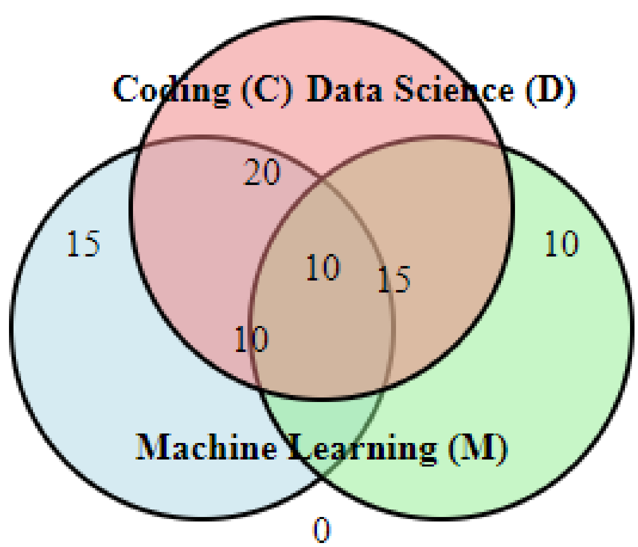
Question no 3:

Step 1: Set up the Venn diagram

- **Given**:
  - $|C \cap D| = 30$
  - $|D \cap M| = 25$
  - $|C \cap M| = 20$
  - $|C \cap D \cap M| = 10$
  - No one attended only Machine Learning.
- **Breakdown**:
  - 20 attended **only Coding and Data Science**.
  - 15 attended **only Data Science and Machine Learning**.
  - 10 attended **only Coding and Machine Learning**.
  - 15 attended **only Coding**.
  - 10 attended **only Data Science**.
- **Total students**:

  15 + 10 + 20 + 15 + 10 + 10 = 80 (total students)

Venn Diagram of Workshop Sessions:



Step 2: Solve for Data Science attendees

- **Coding = Data Science**:
  - Total attendees in **Coding** = 15 + 20 + 10 + 10 = 55
  - Total attendees in **Data Science** = 10 + 20 + 15 + 10 = 55

Thus, the number of students who attended the **Data Science** session is **55**.

Question no 4:

To negate the inequality $\mathbf{0} < \mathbf{x} \leq \sqrt{2}$ using De Morgan's Laws, we follow these steps:

Given Statement:

The inequality $\mathbf{0} < \mathbf{x} \leq \sqrt{2}$ means:

$\mathbf{x} > \mathbf{0}$ and $\mathrm{x} \leq \sqrt{2}$.

Negation Using De Morgan's Laws

The negation of $(\mathbf{0} > \mathbf{x})$ and $(\mathrm{x} \ \leq \sqrt{2})$ is :

$x \leq 0$ or $\mathrm{x} \ > \sqrt{2}$

Final Answer:

The negation of $\mathbf{0} < \mathbf{x} \leq \sqrt{2}$ is: $x \leq 0$ or $\mathrm{x} \ > \sqrt{2}$